

Superstore Sales & Customer Analytics

Dashboard

End -to -End Data Analytics Project



Project Report



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Tools Used: Microsoft Excel, MySQL, Power BI



Dataset Source: Kaggle Superstore Dataset



Date: September-2026

Executive Summary

This project focuses on analyzing retail sales data from a superstore business to uncover valuable insights related to sales performance, profitability, customer behaviour, and product trends.

The analysis was conducted using Microsoft Excel for data preparation, MySQL for business analysis, and Power BI for interactive dashboard development.

The project aims to support data-driven decision-making by identifying high-performing products, profitable customer segments, regional performance trends, and loss-making products.



Business Objectives

The main objectives of this project were:

- Analyze overall sales and profitability.
- Identify top-performing products and categories.
- Evaluate customer purchasing behaviour.
- Compare regional performance.
- Monitor sales trends over time.
- Detect loss-making products.
- Build interactive dashboards for business users.

Dataset Information & Data Cleaning

Attribute	Details
Dataset Name	September Sales Dataset
Source	Kaggle
Records	9994+
Columns	21
Industry	Retail
Tools Used	Excel MySQL Power BI

Key Fields

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

Data Cleaning Process

Data preparation was performed using Microsoft Excel.

- ✓ Activities Performed
- ✓ Verified data quality
- ✓ Checked date formats
- ✓ Removed duplicate records
- ✓ Validated sales and profit values
- ✓ Standardized column structure
- ✓ Prepared dataset for SQL and Power BI analysis

Result

A clean and analysis-ready dataset suitable for reporting and dashboard creation.

SQL Analysis

MySQL was used to perform business analysis and KPI calculations.

Key SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- Aggregate Functions
- CASE Statements
- Window Functions
- Common Table Expressions (CTEs)

Business Questions Answered

Sales Analysis

What is the total sales revenue?

What is the total profit?

What is the average order value?

Category Analysis

Which category generates the highest sales?

Which category generates the highest profit?

Product Analysis

What are the top-selling products?

Which products generate losses?

Customer Analysis

Who are the top customers?

Which customer segment contributes the most revenue?

Regional Analysis

Which region performs best?

Which region has the highest profitability?

Time Analysis

How do sales change over time?

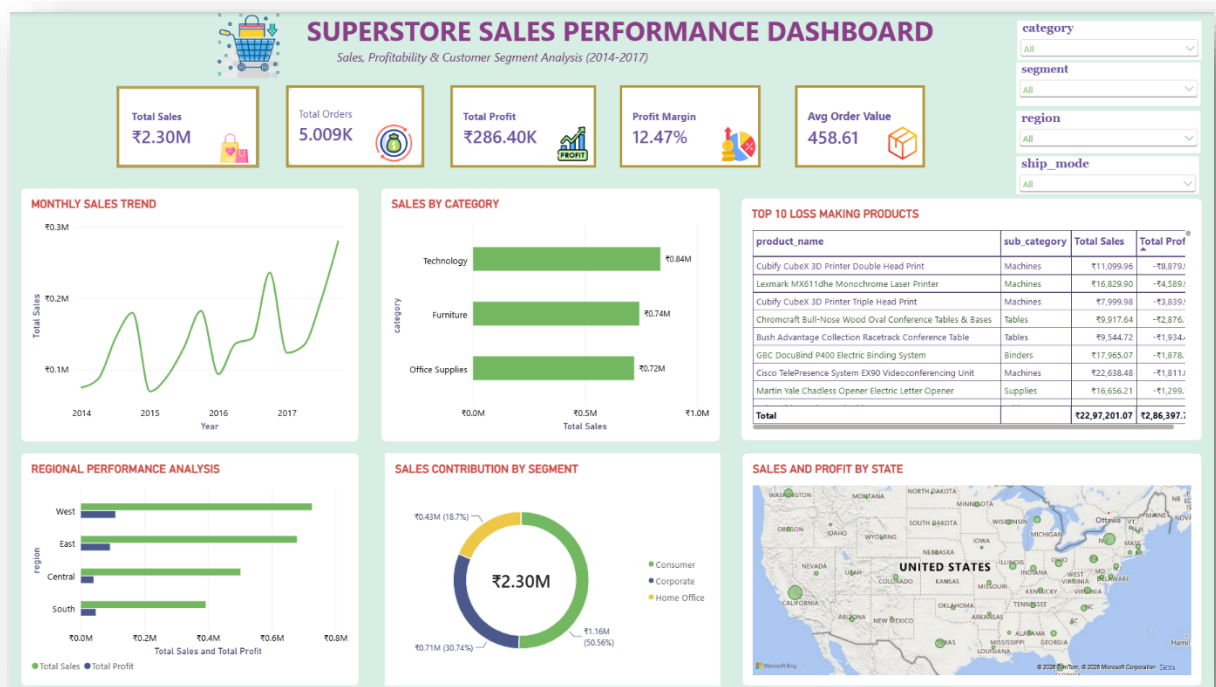
What are the monthly and yearly sales trends?

Dashboard Development

Power BI was used to create interactive business dashboards.

Dashboard 1: Executive Sales Dashboard

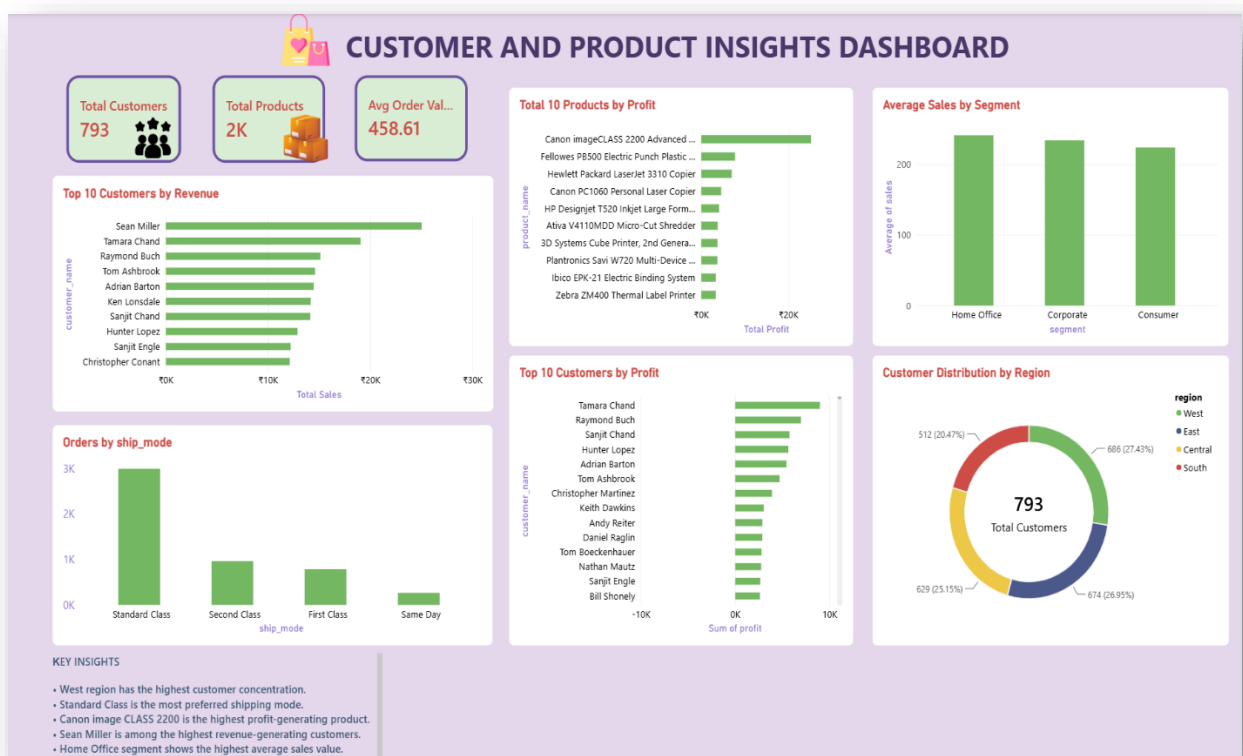
- KPIs
- Total Sales
- Total Profit
- Total Orders
- Profit Margin
- Average Order Value
- Visuals
- Monthly Sales Trend
- Sales by Category
- Sales by Region
- State-wise Sales Map
- Top Loss-Making Products



Dashboard 2: Customer & Product Insights Dashboard

Visuals

- Total Customers
- Total Products
- Average Order Value
- Top 10 Customers by Sales
- Top 10 Products by Profit
- Customer Distribution by Region
- Average Sales by Segment
- Top 10 Customers by Profit
- Order by Ship mode



Key Findings

Sales Insights

Technology category generated the highest sales revenue.

A limited number of products contributed significantly to total sales.

Profit Insights

Certain products generated negative profit despite strong sales performance.

Discounts had a noticeable impact on profitability.

Customer Insights

Consumer segment contributed the largest share of sales.

High-value customers generated a significant portion of revenue.

Regional Insights

The West region demonstrated strong sales performance.

Regional profitability varied across locations.

Business Recommendations

Recommendation 1

Reduce discounts on products that consistently generate losses.

Recommendation 2

Focus marketing efforts on high-performing customer segments.

Recommendation 3

Increase inventory availability for top-selling products.

Recommendation 4

Monitor regional profitability and optimize operations accordingly.

Recommendation 5

Use KPI dashboards regularly to support decision-making.

Skills Demonstrated

Microsoft Excel	Data Cleaning, Data Validation, Pivot Tables
MySQL	Data Analysis, Aggregate Functions, Business KPI Queries, Window Functions, CTEs
Power BI	Data Modeling, DAX Measures, Interactive Dashboards, Data Visualization

Conclusion

This project successfully transformed raw retail sales data into actionable business insights through the use of Excel, MySQL, and Power BI. The dashboards provide a comprehensive view of sales performance, customer behaviour, product profitability, and regional trends, enabling data-driven business decisions.